Oh

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Tous les fichiers svg :

Dedication

For those who hate looking at a template with 500 lines of code and an extra 300 lines commented out.

Declaration

Acknowledgements

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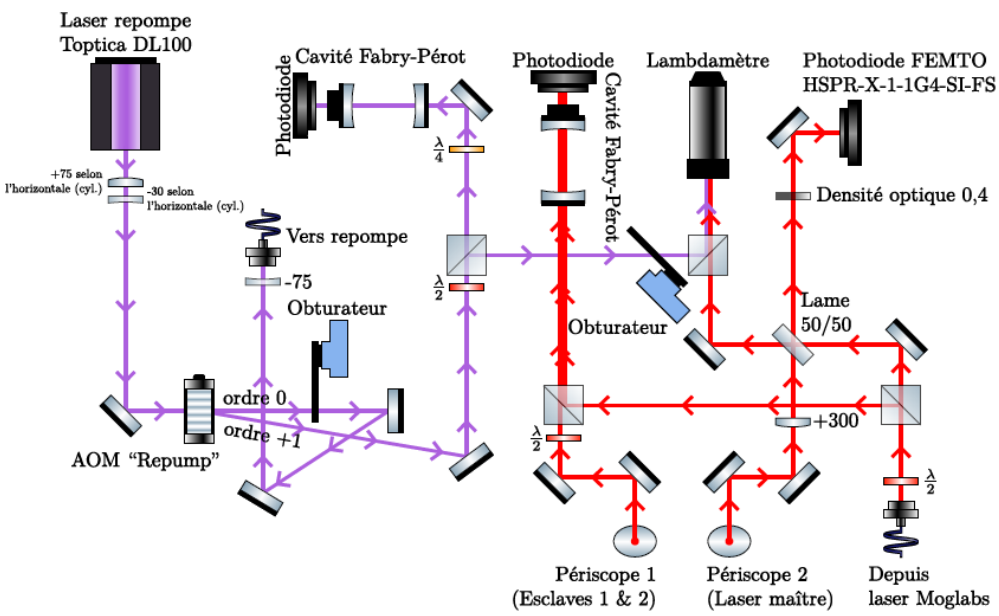

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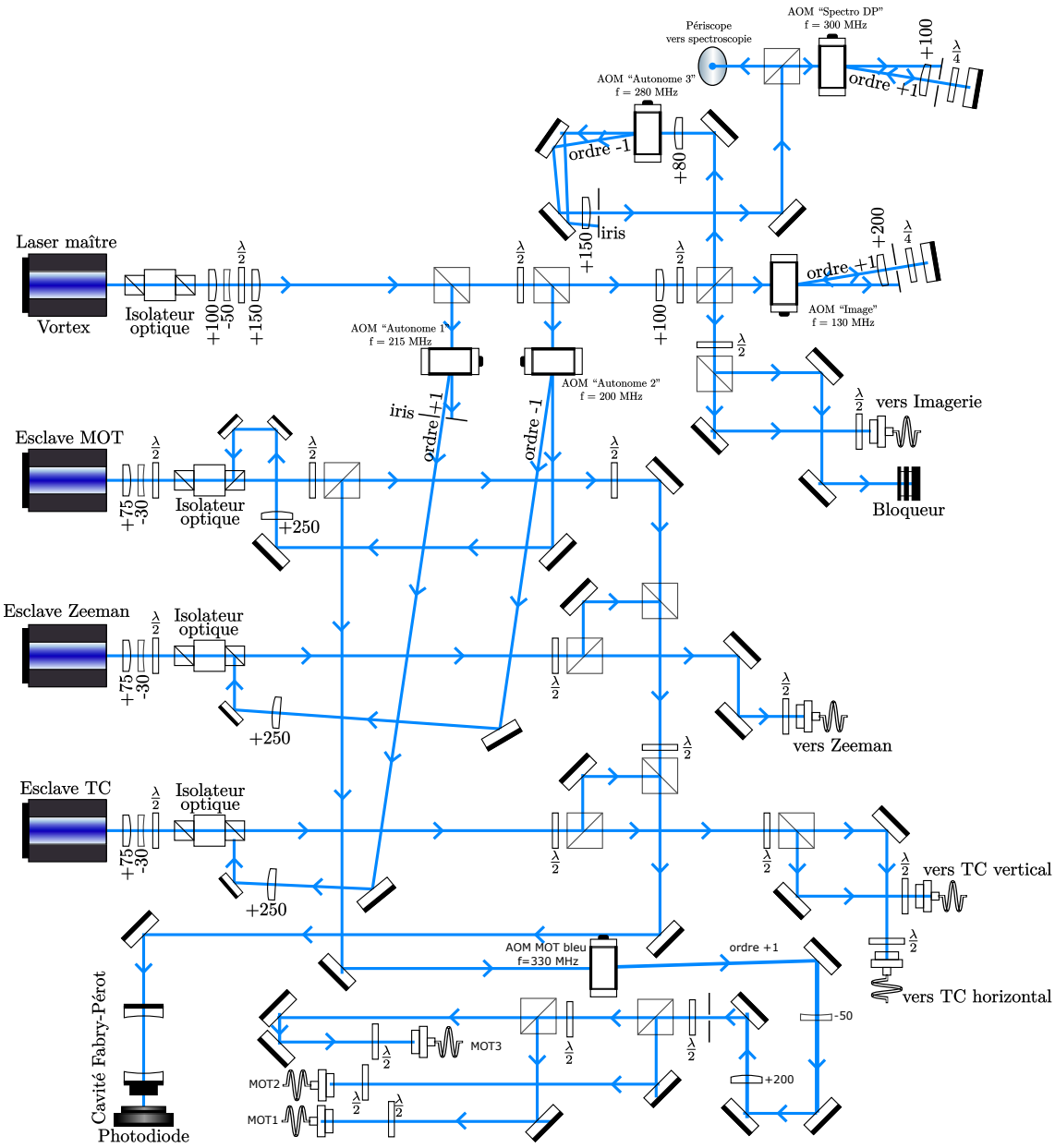

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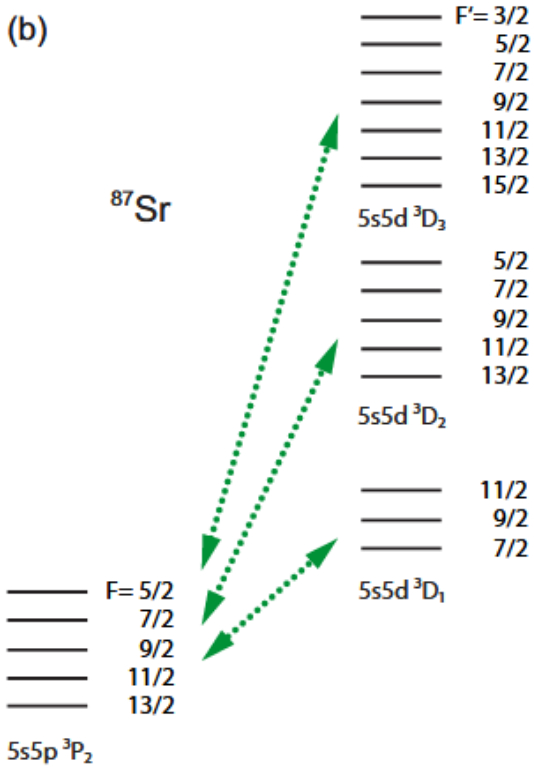

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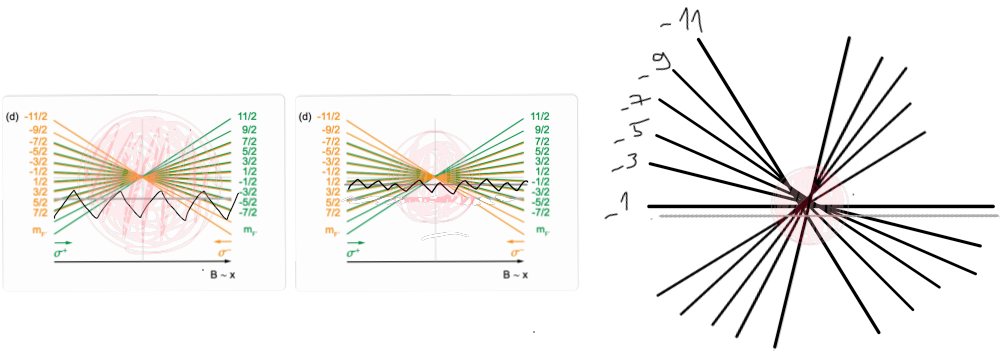

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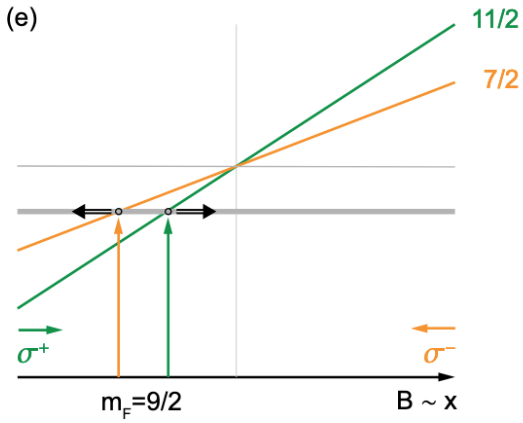
Need a stir because :



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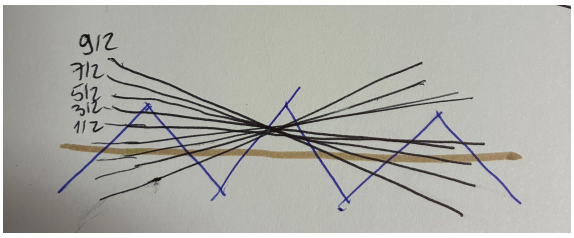

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1.1.7 Narrow MOT

cf p.43 S.Stellmer thesis

1.1.7.1 Optimization of the narrow MOT (intensity, frequency, effect on the size of the cloud)

Include images of the cloud for different I and detuning ?

1.1.7.2 Optical setup

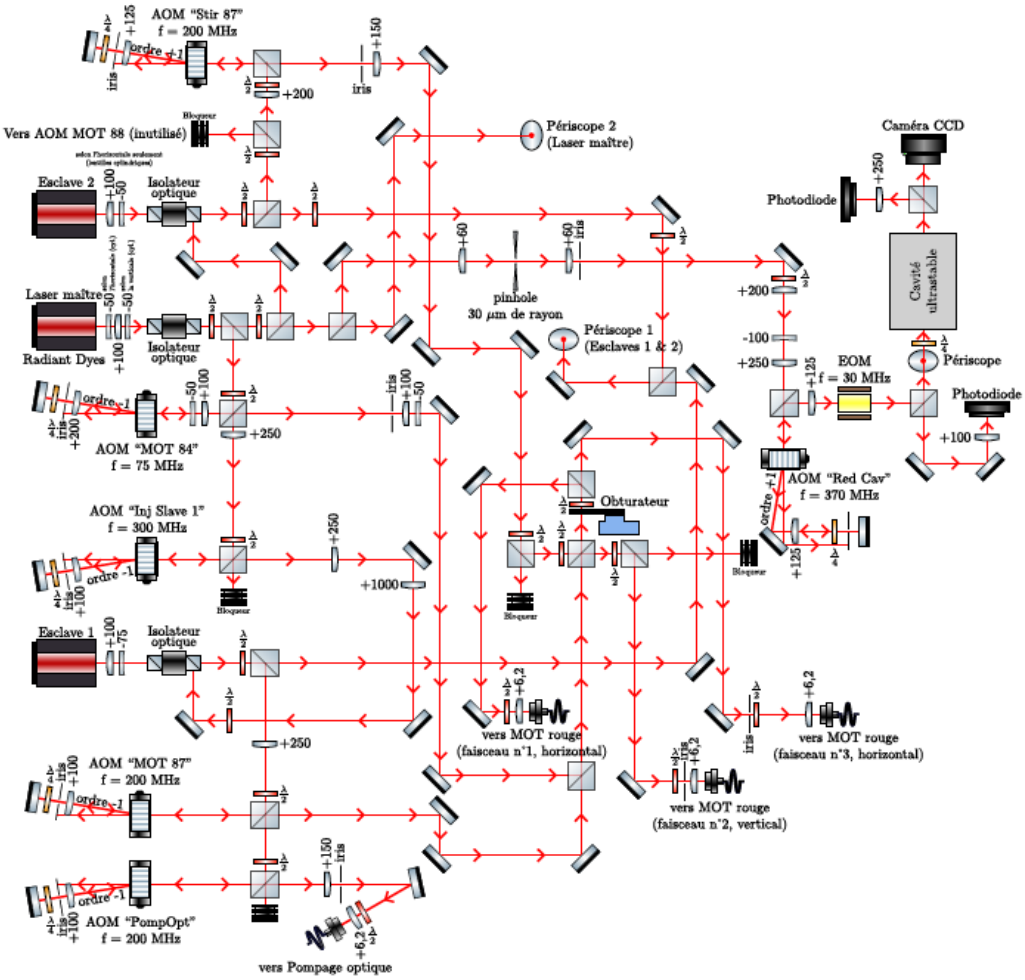

FIGURE 14 – Préparation des différents lasers rouges

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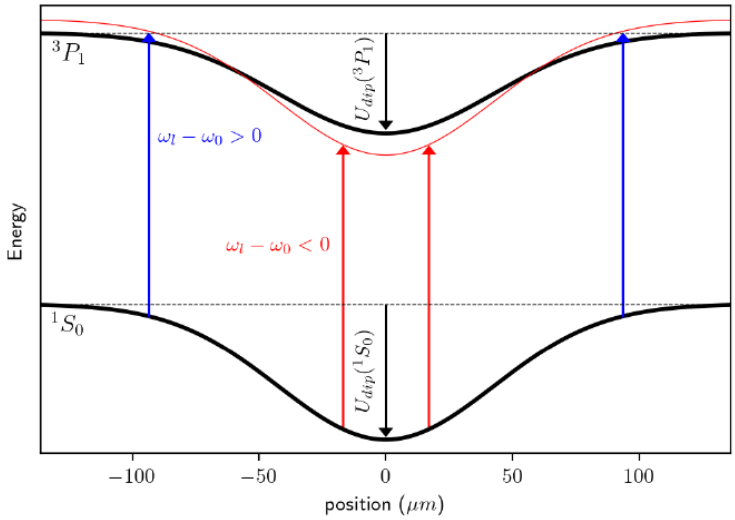

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Comment on the LS it does to each state

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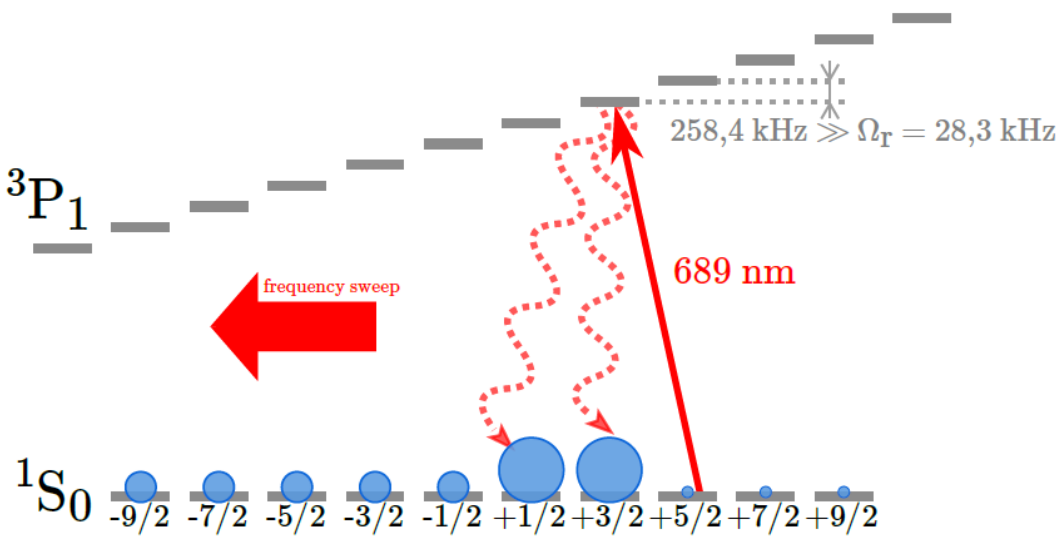

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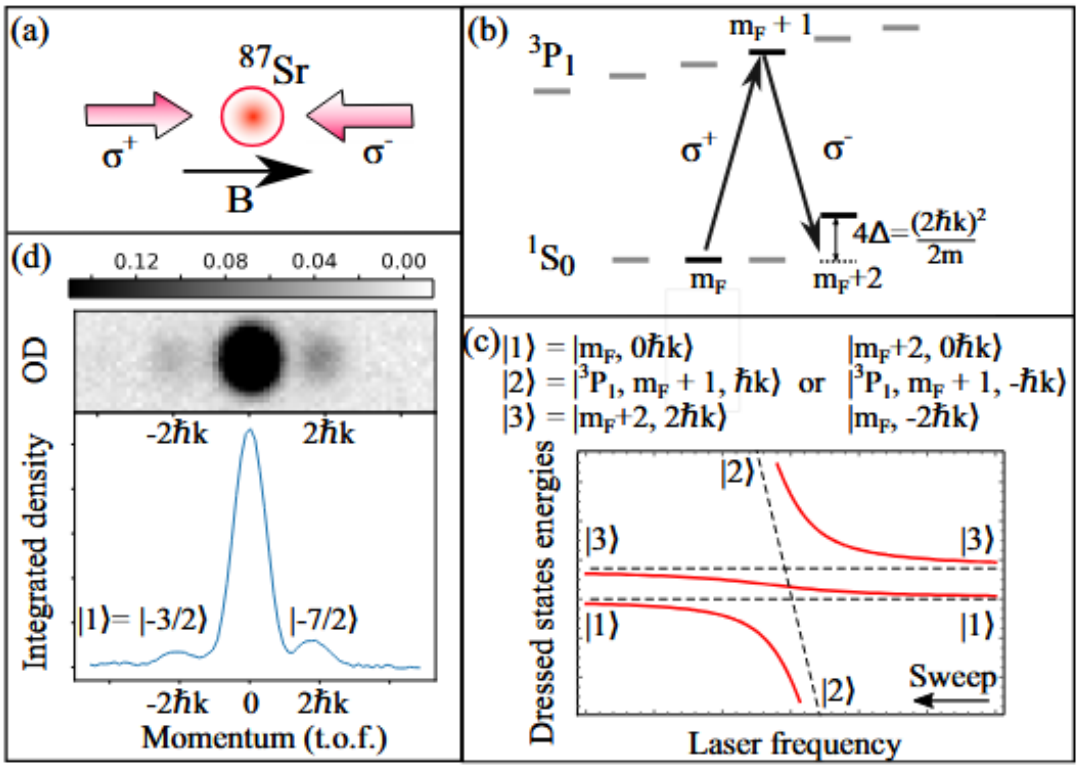

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2.1 Preparation of arbitrary dimension Hilbert space

2.1.1 Raman process

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2.1.1.2 $\delta m_F = \pm 2$

2.1.2 Moglabs chain without cavity

2.1.3 Purification of the laser spectrum with a FP cavity

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2.2 Interferometric sensing with multiple nuclear spin state

2.2.1 Driving long coherence time Rabi oscillations

2.2.1.1 Rabi oscillations

Comment on what the FP could add as a longer coherence time of the qubit

2.2.1.2 Interferometer of su(2) symmetry

2.2.1.3 Discussion on inhomogeneites

2.2.2 Measuring two quantites at a time

2.2.2.1 Physical principle

2.2.2.2 Results

2.2.3 Measuring two non commuting observables

2.2.3.1 Principle

2.3 SU(N) symmetry (ce qu'il faudrait pr la tester e.g densité gaz, alimentation bobines -¿ comment faire mieux que les chiffres actuels)

Chapter 3

Engineering highly entangled system of photoassociated 87Sr atoms

In metrology the current limiting error source of measuring a parameter ϕ is the quantum shot noise $\delta\phi \simeq \frac{1}{\sqrt{N}}$. In the chapter on Ramsey interferometers the phase ϕ acquired by the qubit enhanced by $\frac{\pi}{2}$ pulses was limited by this boundary.
Engineering Dicke states would make possible to go beyond this limitation. For such system variance express as $\delta\phi \simeq \frac{1}{N}$ from the Heisenberg principle being the best accuracy we attempt in metrology. ^{87}Sr accessible large Hilbert spaces enable to improve even more the accuracy of the measurement.
Compléter avec ref [1] et autres

3.1 Introduction on photoassociation

3.1.1 Preamble on scattering theory

In the study of electron-nuclei interactions, the Born-Oppenheimer approximation provides a natural simplification. Due to the large mass ratio $\mu = m_p/m_e \simeq 1000$, the nuclei move on a much slower timescale than the electrons. One can therefore neglect the coupling terms between the nuclear and electronic kinetic energies, and solve the electronic problem on spherical coordinate system centered in nucleus.
Once the nuclear coordinates are fixed, the effective potential experienced by the electrons inherits the spherical symmetry of the problem. This allows the total wavefunction to be factorized into a radial and angular part:

$$\psi(r,\theta,\phi) = R_l(r)Y_l^m(\theta,\phi) \tag{3.1}$$

where the spherical harmonics $Y_l^m(\theta,\phi)$ encode the angular dependance. The Schrödinger equation then reduces to a one-dimensional problem which can be treated for each partial wave l:

$$-\frac{\hbar^2}{2\mu}\left(\frac{1}{r^2}\frac{d^2R_l(r)}{dr^2}r^2 - \frac{l(l+1)}{r^2} + \hat{V}(r)\right)R_l(r) = ER_l(r) \tag{3.2}$$

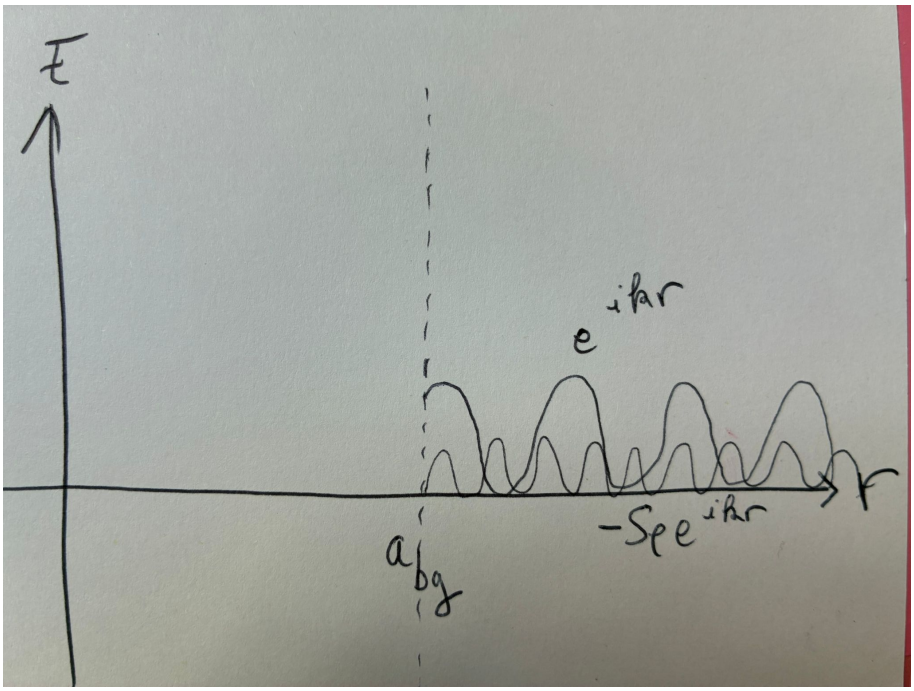


Figure 3.1: Caption

This one-dimensional structure allows an analogy with the Landau-Zener framework, as will be discussed below. To extract the scattering properties of the system, one studies the asymptotic behavior of the solutions to this equation. At large distances, where the potential $\hat{V}(r)$ vanishes, the general solution for each partial wave l takes the form [2]:

$$R_l(r) \xrightarrow{r\to\infty} \frac{a_l}{r} sin(kr - \frac{l\pi}{2} + \delta_l) \tag{3.3}$$

where a_l is a normalization factor and δ_l the phase acquired by the l-th partial wave due to the interaction potential.

The full wavefunction is recovered by summing over all partial waves. At large distances, this superposition can be rewritten as an incoming plane wave and an outgoing scattered spherical wave:

$$\Psi \xrightarrow{r\to\infty} e^{i\mathbf{k}\cdot\mathbf{z}} + f(\theta)\frac{e^{ikr}}{r} \tag{3.4}$$

where $f(\theta)$ is the scattering amplitude which determines wheter the collision is elastic or inelastic and θ is the angle between the incident and scattered wavevectors.

Requiring continuity of the wavefunction across all regions imposes a constraint on the phase shifts. At low momenta k one finds:

$$\delta_l \propto k^{2l+1} \tag{3.5}$$

For s-waves ($l = 0$), this leads to the definition of the scattering length:

$$a = -\lim_{k \to 0} \delta_0 / k \tag{3.6}$$

which characterizes the effective range of the interaction in the low-energy limit. To further understand how the atoms scatter, the scattered part $\Psi - e^{ikz}$ on 3.4 is expressed in a spherical basis as a sum over all the partial waves:

$$\Psi \simeq \frac{i}{2kr} \sum_{l=0}^{\infty} (2l+1) \left((-1)^l e^{-ikr} - S_l e^{ikr} \right) \tag{3.7}$$

with $S_l = e^{2i\delta_l}$ the scattering amplitude.

For a **conservative hamiltonian** $|S_l| = 1$ meaning $\delta_l \in \mathbb{R}$. In this case the wavefunction is well a sum of incoming and outcoming waves of same amplitude.

The elastic cross section representing the surface where the scope of the potential is effective writes as

$$\sigma_{el} = \frac{\pi}{k^2} \sum_{l=0}^{\infty} (2l+1) |1 - S_l|^2 \tag{3.8}$$

But we see in equation 3.7 that $|S_l|$ needs to be < 1 to allow dissipation of the wavefunction. Defining δ_l as a complex number satisfies this condition.

In this case the inelastic collisions cross section reads as

$$\sigma_{inel} = \frac{\pi}{k^2} \sum_{l=0}^{\infty} (2l+1)(1 - |S_l|^2) \tag{3.9}$$

In ultracold gases, only s-wave atoms can collide because they do not have enough kinetic energy to pass the potential barrier of higher angular momentum states so $l = 0$ and the scattering amplitude is

$$S_0 = e^{2i\delta_0} \simeq 1 + 2i\delta_0 \tag{3.10}$$

By using the result of 3.6

$$1 + 2i\delta_0 \simeq 1 - 2ika \tag{3.11}$$

Because δ_0 is complex, a direct result of these equations is that we can write a general scattering length as

$$a = a_{bg} + ia" \tag{3.12}$$

At low temperatures we can re-express equations 3.8 and 3.9 as

$$\sigma_{el} = 4\pi |a|^2 \tag{3.13}$$

$$\sigma_{inel} = 4\pi|a"|^2 \tag{3.14}$$

A usual case of the elastic collisions is atoms in the ground state. For inelasticity photoassociation is a good example of dissipative process where two atoms in the ground state are coupled to a molecular state by a laser.

3.1.2 What is photoassociation

Photoassociation (PA) is a process from which free atoms pair our bounded to each other through two-body and three-body losses via a resonant laser.
In the basis $\{|\psi_g\rangle, |\psi_e\rangle\}$ of free atoms in a radial potential $V_g(r)$ and first molecular state $V_e(r)$ in the absence of the resonant laser the equation evolution of the system express as

$$i\hbar\frac{d}{dt}\begin{pmatrix}\psi_g\\ \psi_e\end{pmatrix} = \begin{pmatrix}-\frac{\hbar^2}{2\mu}\nabla + V_g & 0\\ 0 & -\frac{\hbar^2}{2\mu}\nabla + V_e\end{pmatrix}\begin{pmatrix}\psi_g\\ \psi_e\end{pmatrix}. \tag{3.15}$$

Then a resonant laser is tuned to an internal state of $V_e(r)$ to bound the atoms pair as presented in figure 3.2

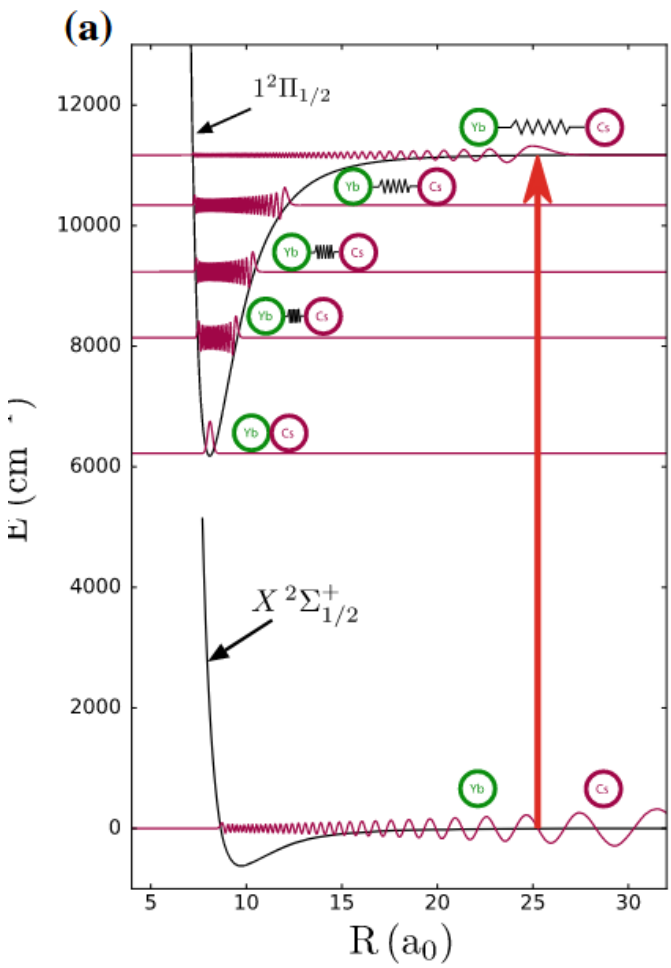


Figure 3.2: Caption

This operation consists in adding inelastic collisions to the system. Then the effective evolution equation of the coupled atoms is

$$i\hbar\frac{d}{dt}\begin{pmatrix}\psi_g\\ \psi_e\end{pmatrix} = \begin{pmatrix}-\frac{\hbar^2}{2\mu}\nabla + V_g & \Omega/2\\ \Omega/2 & -\frac{\hbar^2}{2\mu}\nabla + V_e + \hbar\Delta - i\frac{\gamma_{mol}}{2}\end{pmatrix}\begin{pmatrix}\psi_g\\ \psi_e\end{pmatrix}. \tag{3.16}$$

with γ_{mol} the molecular spontaneous emission linewidth.

Then sum of partial waves determined in 3.7. are solutions of this hamiltonian. We can deduce the phase δ_l (see 3.3) and so the scattering length are then modified. The real part of a encodes two-body losses rate whereas its imaginary part changes the collisions properties. In this case we can write an effective scattering length enclosing the background scattering length a_{bg} standing for elastic collisions without any coupling, δa a modification of it because of the coupling and still the inelastic part of it

$$a = a_{bg} + \delta a + ia" \tag{3.17}$$

One can express the atom number evolution depending on this linewidth as

$$\frac{dN}{dt} = -\frac{\beta N^2}{V_e} - \Gamma N \tag{3.18}$$

where V_e is the effective volume of the cloud $(2\pi)^{3/2}\sigma_x\sigma_y\sigma_z$ for a density Gaussian distribution, Γ the one-body loss rate limited by vacuum quality and 2-body losses express as $\beta \propto \gamma_{mol}$ and is possible to determine by a fit.

Finally in a context of spectroscopy, to adress more bound states one sweep the laser frequency from the free-atomic line to red detuned frequencies. The quantity that will testify there is two-body losses is the atom number remaining in the trap after a photoassociation pulse. The molecule leaves the trap by acquiring kinetic energy from the binding energy.

PA is mostly used in Feschbach resonance field because determining the exact position of the vibrational states enables by changing the magnetic field to tune the scattering length of the atoms and thus the interactions in the system.

3.1.3 Parameters for PA study

Many experimental parameters enable to maximize the two-body loss rate of the photoassociated atoms. This rate K_2 introduced above can express as

$$K_2 = \frac{\beta}{V_e} = \frac{1}{k}\frac{\frac{1}{4}\gamma_{stim}\gamma_{mol}}{\Delta^2 + (\frac{1}{2}(\gamma_{stim} + \gamma_{mol})^2)} \tag{3.19}$$

To understand what governs K_2, we need to look more closely at γ_{stim} which encodes the coupling between the colliding atom pair and the PA laser. This rate is linear

with the laser intensity, proportional to the 3D density n, and to the probability of transitioning from $|\psi_g\rangle$ to $|\psi_e\rangle$ the Franck-Condon factor:

$$F_{FC} = \left| \int_0^\infty \psi_g(r)\psi_e(r)dr \right|^2 \tag{3.20}$$

This factor directly measures the overlap between the initial and final state wavefunctions, and therefore quantifies how efficiently the PA laser can drive the transition. The Condon radius is the internuclear distance at which this overlap is maximum, meaning in a classical picture this is where the atoms spend the most time — and thus where the photoassociation is most likely to occur.
Since γ_{stim} depends on both the laser intensity and the density, these become the key experimental knobs to tune K_2. To make this more concrete and allow comparison with other experiments, the photoassociation rate is commonly re-expressed in the molecular community through an optical length:

$$l_{opt} = \frac{\beta m}{16h} \tag{3.21}$$

m the atomic mass of the atom. This single quantity captures the overall PA efficiency, and from this formula it is clear that the main experimental levers to increase the PA rate are the cloud density and the PA beam intensity — precisely what we will test in the experimental results section.

From loss rate to molecular structure
So far we have described the PA process from the perspective of atom losses. But the molecules formed in this process have a rich internal structure that determines which transitions are accessible and how efficiently they can be driven — in particular through the Franck-Condon factor. Understanding this structure is therefore essential to optimise K_2 experimentally.

A first key property is the symmetry of the orbital wavefunction. For an even wavefunction the exchange of the two atoms A and B does not change the sign of the wavefunction, as in left figures 3.3. The potential is then qualified **gerade**. The probability of finding electrons in between the two atoms is high compared to the **ungerade** potential where the odd wavefunction cancels the probability of finding electrons in between.

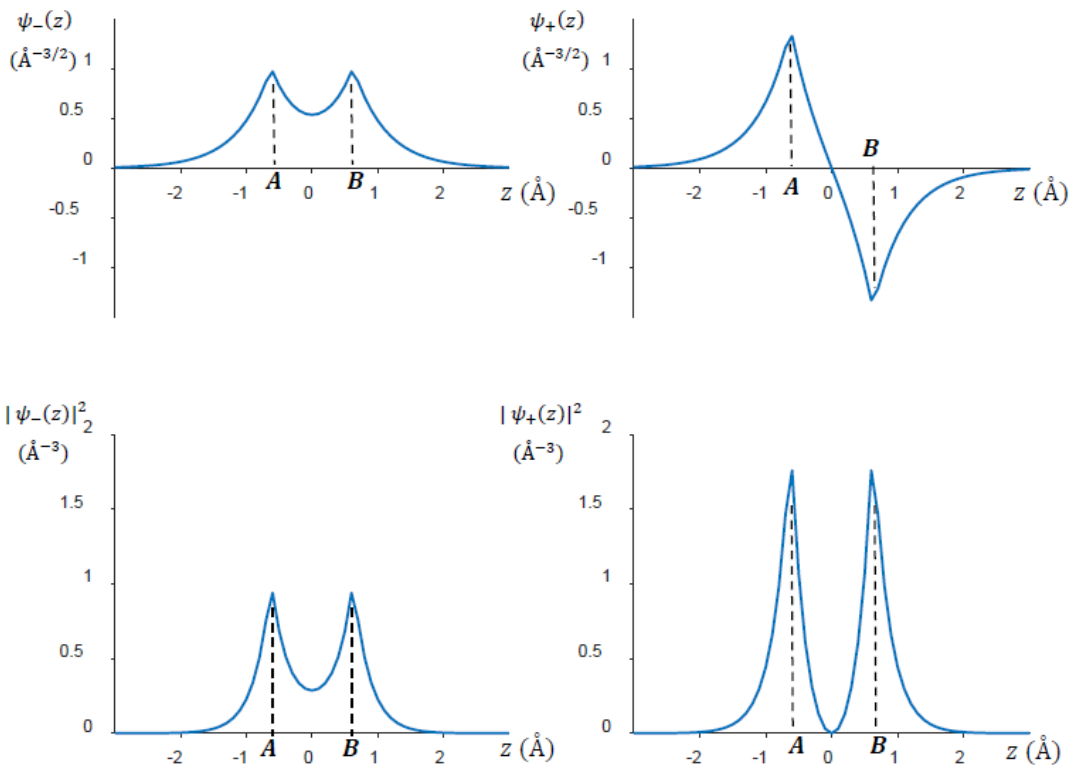


Figure 8.5 : Evolution le long de l'axe internucléaire Oz ($x=0, y=0$) des fonctions d'onde liante $|\psi_-\rangle$ et antiliante $|\psi_+\rangle$ et des densités de probabilité de présence de l'électron associées $|\psi_-|^2$ et $|\psi_+|^2$. La distance internucléaire est fixée à $R = 1{,}25$ Å (minimum de E_-).

Figure 3.3: u-g symmetry

While exciting a single atom is straightforward, identifying the correct molecular state to target is considerably more involved, since a large number of potentials and internal states coexist. To navigate this complexity, we now introduce the molecular quantum numbers that label these states and determine their accessibility.

The good quantum numbers

The total angular momentum of the molecule is $T = R + F = R + I + J$, F being the total spin of the two atoms $f_1 + f_2$, $J = j_1 + j_2$ encodes the spin-orbit coupling, and R is the rotational angular momentum of the molecule. The projection of the angular momenta onto the internuclear axis is:

$$\Omega = \Lambda + \Sigma \tag{3.22}$$

where Λ is the projection of the orbital momentum of the molecule on the internuclear axis and Σ the projection of the momentum spin. M_T denotes the projection of T onto a chosen quantization axis.

For the rotational quantum number there is a parity of the total wavefunction that needs to be fulfilled. In addition there are only s-wave collisions in cold gases then the part of the wavefunction should be odd the spin one antisymetric

$$|\Psi\rangle = \left(|\phi_e\rangle_A |\phi_g\rangle_B + (-1)^R |\phi_e\rangle_B |\phi_g\rangle_A\right) \otimes |\chi\rangle \tag{3.23}$$

$|\phi_e\rangle$ and $|\phi_g\rangle$ are the electronic wavefunctions of the two atoms A and B in ground and excited state respectively, $|\chi\rangle$ is the spin wavefunction. This restricts the accessible values $R = 0, 2$ or 4... which directly limits the number of rotational states that can be populated during photoassociation.

For $^1S_0$ - $^3P_1$ molecular state there is a strong spin-orbit coupling that couples its related momentum to the internuclear axis (3.4), quantified by Ω and described as Hund's case (c)

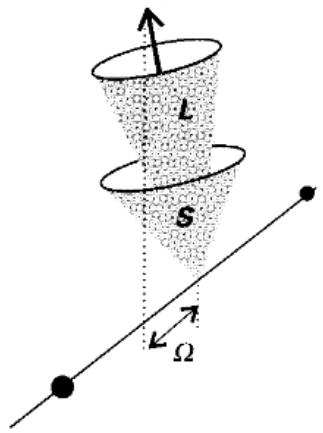


Fig. 11.4 A vector diagram for Hund's case (c), in which the spin–orbit coupling is very strong.

Figure 3.4: hund

The experiment presented in the following section is photoassociation in the $F = 11/2$ hyperfine state. We will focus in this state to simplify the discussion.
Say we have one atome in the $^1S_0$ state and one in the $^3P_1$ state: we can have a total spin of $J = 0, 1$ or 2. It gives possible values of total atomic angular momentum F:

$$\text{For } J = 0, \quad F = \frac{9}{2} \tag{3.24}$$

$$\text{For } J = 1, \quad F = \frac{7}{2}, \frac{9}{2}, \frac{11}{2} \tag{3.25}$$

$$\text{For } J = 2, \quad F = \frac{5}{2}, \frac{7}{2}, \frac{9}{2}, \frac{11}{2}, \frac{13}{2} \tag{3.26}$$

We have one atom with $f_1 = 9/2$ from the $^1S_0$ state and one in the $^3P_1$ state with

$f_2 = 11/2$. The possible values of total angular momentum of the molecule are

$$\text{For } R = 0, \quad T = 1, 2, 3...10 \tag{3.27}$$
$$\text{For } R = 2, \quad T = 1, 2...12 \tag{3.28}$$
$$\text{For } R = 4, \quad T = 1, 2...14 \tag{3.29}$$

The initial $T^i = i_1 + i_2$ with $R = 0$ because we are in s-wave collisions. In respect with the parity $p = +1$ the possible values of T^i are $T^i = 0, 2, 4....$
As for atoms there are selection rules for transitions from a free state to a molecular state fixed by the angular momentum of the electrons. $\Delta J = 0, 1$ gives $\Delta T = 0, 1$ with $T = 0 \rightarrow T = 0$ forbidden.
In the end the accessible transitions in the first molecular excited state of the $f_1 = 11/2$ are

$$\text{For } R = 0, \quad T^f = 1, 2, 3...11 \tag{3.30}$$
$$\text{For } R = 2, \quad T^f = 1, 2...13 \tag{3.31}$$
$$\text{For } R = 4, \quad T^f = 1, 2...15 \tag{3.32}$$
$$... \tag{3.33}$$

There are 981 combinations of accessible (R, T^f) molecular external states and for each potential a numerous internal state we are going to demonstrate briefly in the following.

3.1.4 Internal energy states

3.1.4.1 WKB approximation

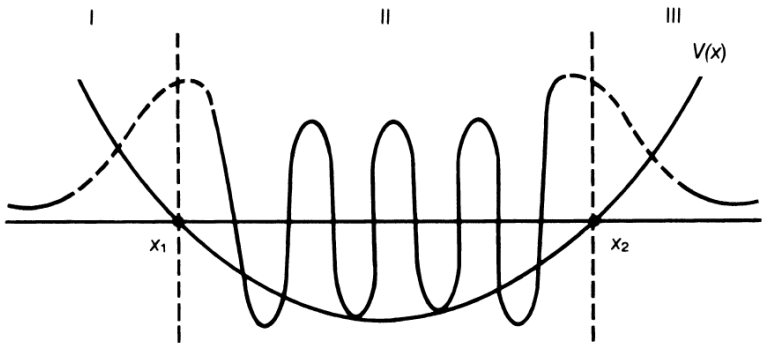
WKB approximation is a semi-classical approach where plane waves $\psi(x) = \frac{C}{\sqrt{(k(x))}} e^{i \int_0^\infty k(x) dx}$ are solutions of the 1D Schrödinger equation for two particles of eigenenergie E in a potential $V(x)$

$$\frac{-\hbar^2}{2\mu} \frac{d^2\psi}{dx^2} + V(x)\psi = E\psi \tag{3.34}$$

giving the expression of the wavevector k

$$k = \frac{p}{\hbar} = \sqrt{2\mu(E - V(x))} \tag{3.35}$$

The condition of validity $\frac{d\lambda}{dx} \ll 1$ tells that the atoms must follow adiabatically the potential to adapt their movement to its variation. It means at really small x of region I of figure 3.5 this approximation is not valid.



IGURE 2.1. Schematic diagram for behavior of wave function $u_E(x)$ in potential well $V(x)$ ith turning points x_1 and x_2.

Figure 3.5: Caption

At position $x = x_1$ and x_2 the momentum diverges which means that the plane wave approximation is not valid anymore. In the region I and III the wave is evanescent with a phase $k = i\sqrt{2\mu(|E - V(x)|)}$. It is then usual the average the potential as linear with x around these turning points.

By reflection with the left barrier of the potential at $x = x_1$ the stationnary wave must come back with a 2π at x_2. By continuity of the solutions in $[0, +\infty[$ the phase tends to

$$\int_{x_1}^{x_2} k(x)dx = (n + \frac{1}{2})\pi \tag{3.36}$$

Then this formula enables to identify the solutions E that satisfies the condition $(n + \frac{1}{2})\pi$. These values are what we call the vibrationnal states of a molecular potential.

3.1.4.2 Leroy-Bernstein theory

Leroy-Bernstein theory enables to identify the position of the less bounded states without knowing the behaviour of the potential that could be complicated to estimate and solve numerically. This model is valid only close to the dissociation limit occuring at long distance. Then we can approximate the potential as

$$V(r) \xrightarrow[r\to\infty]{} D_e - \frac{C_n}{r^n} \tag{3.37}$$

with D_e the dissociation energy. For a vibrational energy state ν of energy E_ν we express the energy difference $\epsilon = D_e - E_\nu$ to inject it in 3.36 and find solutions of

$$\int_{x_1}^{x_2} \sqrt{2\mu(\frac{C_n}{r^n} - \epsilon)}dr = (\nu_D - \nu + \frac{1}{2})\hbar\pi \tag{3.38}$$

with ν_D the dissociation number. After few calculations we finally obtain

$$\epsilon = \left((\nu_D - \nu) \frac{\sqrt{\pi}\hbar\nu}{\sqrt{2\mu C_n^{2/n}}} \frac{\Gamma(\frac{3}{2} - \frac{1}{\nu})}{\Gamma(1 - \frac{1}{\nu})} \right)^{\frac{2\nu}{\nu-2}} \tag{3.39}$$

Γ being the Beta Euler functions. This energy also writes as

$$\frac{E_\nu - D_e}{\nu_D - \nu} = \left(\frac{\sqrt{\pi}\hbar\nu}{\sqrt{2\mu C_n^{2/n}}} \frac{\Gamma(\frac{3}{2} - \frac{1}{\nu})}{\Gamma(1 - \frac{1}{\nu})} \right)^{\frac{2\nu}{\nu-2}} \tag{3.40}$$

from which we determine the energy difference between the dissociation limit and one of the least bounded states ν of the molecule.

3.1.5 External energy states

As seen in the subsection 3.1.1 in a molecular picture the hamiltonian express as

$$\hat{H} = \frac{\hat{p_r}^2}{2\mu} + \frac{\hbar^2}{2\mu r^2} R\,(R+1) + \hat{V}_{BO} + \hat{H}_{HF} \tag{3.41}$$

It is possible to neglect some orders of electronic interactions depending on the Hund's case and the distance region between the atoms. In our case we approximate the potential to be as for ^{173}Yb in [3]

$$\hat{V}_{BO} = \frac{-C_6}{r^6}\left(1 - \frac{\sigma^6}{r^6}\right) - s\frac{C_3^\Omega}{r^3} \tag{3.42}$$

$$\hat{H}_{\text{hf}} = A\,(\mathbf{i}_1 \cdot \mathbf{j}_1) + B\,\frac{3(\mathbf{i}_1 \cdot \mathbf{j}_1)^2 + \frac{3}{2}(\mathbf{i}_1 \cdot \mathbf{j}_1) - i_1(i_1+1)\,j_1(j_1+1)}{2i_1(2i_1-1)\,2j_1(2j_1-1)} \tag{3.43}$$

$s = \pm 1$ for gerade and ungerade potentials respectively. The hyperfine term of the atom in $^1S_0$ is zero because its electronic angular momentum is zero. Only the hyperfine interaction of the atom with i_1, j_1 in $^3P_1$ counts.

3.2 About photoassociation on other species

3.2.1 Mass scaling (^{88}Sr)

^{88}Sr is a bosonic isotope of strontium with no hyperfine structure. The photoassociation spectrum is then simpler than the fermionic ones : only two potentials in the first excited molecular state 0_u and 1_u. Results have been presented in [4] where their fitting model gives the following potential expression

$$\hat{V}_{BO} \sim \frac{-C_{6,\Lambda}}{r^6} - \frac{C_3}{r^3} + \frac{C_J}{r^2} \tag{3.44}$$

for the two accessible potentials $\Lambda = 0, 1$.

Mass scaling method presented in [5] enables to predict the position of the vibrational states of the different isotopes of strontium. The authors consider the potential is the same for all the isotopes because we can validate the Born-Oppenheimer estimation for all isotopes. It is actually wrong to make this approximation in the case of 87^Sr because the hyperfine structure modifies the expression of the potential. It is still giving an average position of the vibrational states. Then for this method only the reduced mass in the integral

$$\int_{x_1}^{x_2} \sqrt{2\mu(E - V(x))}dx = (n + \frac{1}{2})\pi \tag{3.45}$$

will change upon the isotope. Then with respect with the condition on the phase the energies E will be modified. The experimental positions of the first vibrationnal states of the O_u^+ potential for ^{84}Sr, ^{86}Sr and ^{88}Sr are represented by the red crosses in figure 3.6 and enables to adjust $V(r)$ shape. In blue lines the fitted functions are plotted from this model : it is then possible to deduct the position of the ^{87}Sr vibrationnal states thanks to the other isotopes lines.
We see discontinuites at the 84 isotope: these are avoided crossing coming from a coupling with the ground state potential.

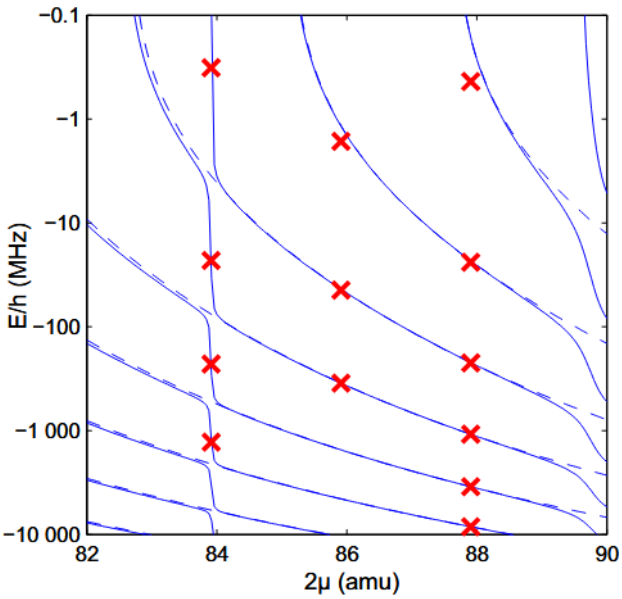


Figure 3.6: Caption

3.2.2 Ytterbium: hfs

Whereas the Strontium bosonic isotope ^{88}Sr has only two molecular channel 0_u and 1_u on the intercombination line it is more convenient to refer to species with a hyperfine structure such as ^{173}Yb. Its molecular potential landscape is more complex (see figure 3.7) than many species that have been studied and the density of each momentum projection $0_u, 1_g, 1_u, 0_g$ comes from the hyperfine structure. It is indeed

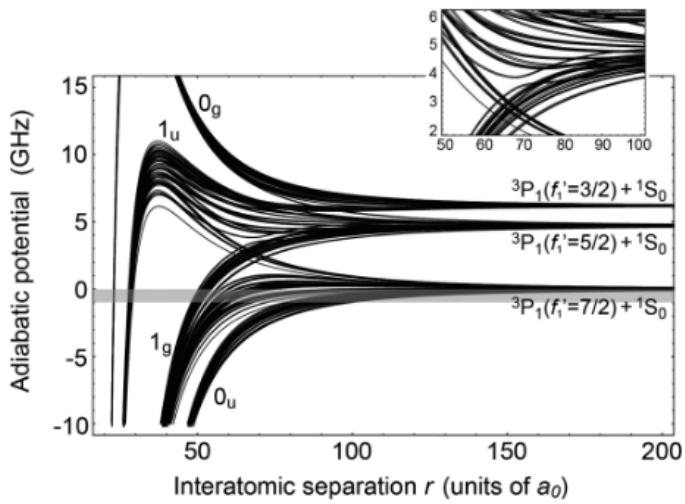


FIG. 2. Adiabatic molecular potentials for a $^{173}Yb_2$ dimer in the $^1S_0 + {}^3P_1$ channel as functions of the interatomic separation r. The molecular potentials for 205 different (T, F, R) configurations are displayed, which are accessible via PA from the initial s-wave colliding atoms in the $^1S_0 + {}^1S_0$ channel. At large r, the potentials converge to three asymptotic branches which correspond to excited atomic states with hyperfine numbers of $f_1' = 3/2$, $5/2$, and $7/2$. Some of the potentials have a local minimum (inset), possibly hosting purely long-range bound states [14]. The energy offset is adjusted to the $f_1' = 7/2$ asymptote. The shaded region indicates the spectral range of our measurements.

Figure 3.7: Caption

expected to count 205 different configurations of (T, F, R) for the excited molecular states on intercombination line with an atomic potential written as 3.42.

Around 80 lines were measured. Atom loss spectra as a function of PA laser detuning reveal the position of the bound states relative to the atomic resonance $|^1S_0, F = 5/2\rangle \rightarrow |^3P_1, F = 7/2\rangle$. Near this resonance atoms are illuminated for 30ms at low intensity $I = 37\text{mW/cm}^2$ to avoid power broadening, allowing the nearest bound state to the dissociation to be resolved. At larger red detunings the intensity is increased and the illumination time optimized to maximize contrast.

3.3 Protocol for PA process

Dicke states are particularly interesting to engineer for metrology because it is possible to go beyond quantum shot-noise limit on a specific parameter ϕ. The strategy we performed to attempt this high precision is to photoassociate atoms to induce two-body losses resulting in single atoms entangled to the $N-1$ other. The aim is to induce Dicke state on a gas of fermions with a precision that goes beyond quantum shot-noise limit.
As introduced in the preamble on scattering theory only s-wave collisions occur in

ultracold gases. It enforce that for fermions -by parity of the total wavefunction being antisymetric- only atoms with even angular momentum can collide meaning for PA process their spin state must be antisymetric so is the case of ^{87}Sr. For bosons it is the contrary: only atoms in triplet states will be photoassociated by symetry of the total wavefunction.
Prior to the main experiment on ^{87}Sr, we calibrated our protocol on the ^{88}Sr isotope, whose photoassociation spectrum is well established in the literature [6], [5], [4].

PA rate highly depending on cloud density as discussed in 3.1.3 the first step for Dicke state engineering is to produce highly dense cloud managed differently depending on the fermionic or bosonic nature.
For fermions the convention to produce dense cloud is to optimize the evaporation ramps on a cloud already well charged in the ODT crossing.
The near-vanishing scattering length of bosonic strontium 88 is detrimental to evaporative cooling, as it severely limits the elastic collision rate required to thermalize the gas. Then only the previous cooling and trapping step count on the density improvement.

Once we have a dense cloud the PA process starts by making the atoms interact with a beam of size $w_0 =$ we tune the frequency by an ensemble of AOMs detailed in the next subsection. From the atomic reference $^3P_1, F = 11/2$ we change the RF frequency delivered by a DDS9852 of the demodulating signal used in beating lock. Doubler frequency module commit to adress lines down to 600MHz from the atomic resonance.
Having find loss peaks with constrast better than the rms value of base cloud the measurement of atoms loss at fixed frequency but various illumination time verifies 3.18.

3.3.1 Setup

Changing the locking system of the MOGlabs enabled us to adress resonances 1GHz under the frequencies previously targeted in Chapter 2. This procedure conserves the set-up presented in **??** so it still serves for Raman manipulation or PA.
While we were injecting a fiber at the output of the "AOM PompOpt" that was directly sent to the vacuum cell the red chain at this level has been modified. The tuned beam assures then the optical pumping and MOGlabs beating 1GHz under . This procedure

This frequency is tuned by the successive AOMs "Inj slave 1" and "PompOpt"

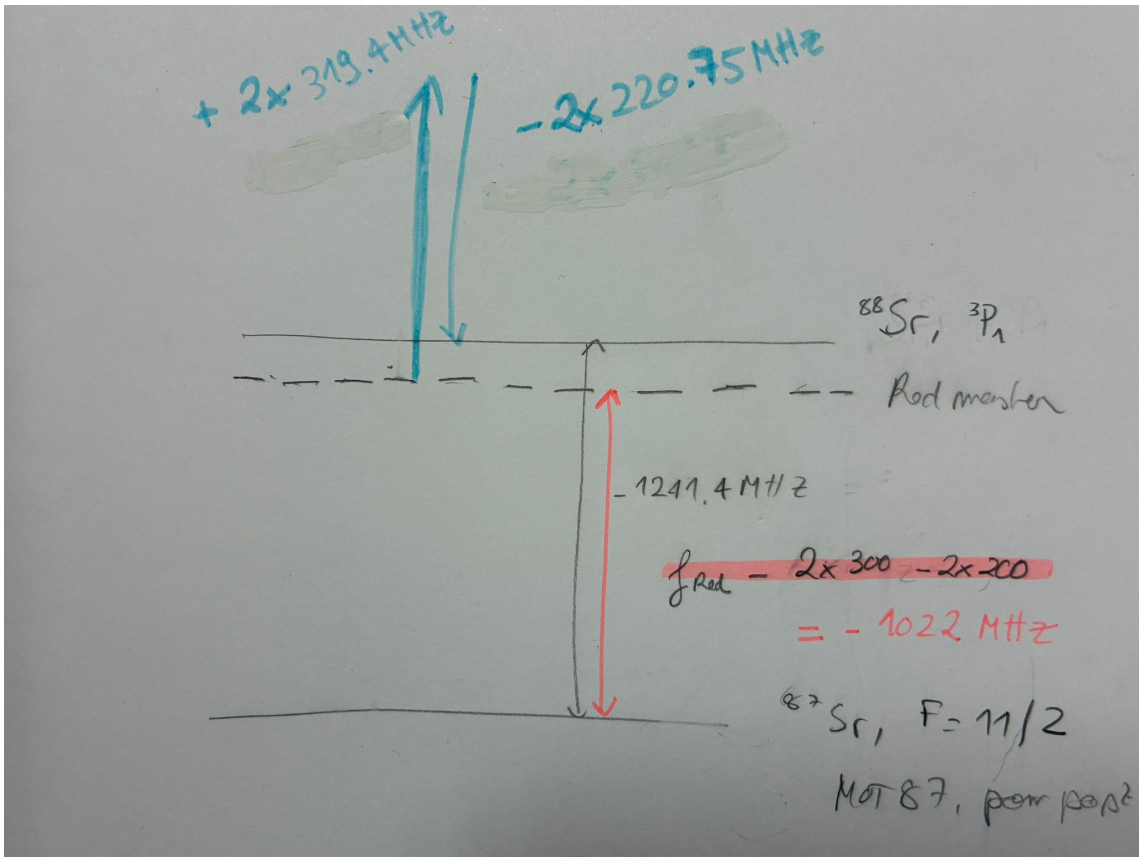


Figure 3.8: Caption

3.4 88Sr Results

For PA process on strontium 88 the beam line currently used for strontium 87 red MOT and 11/2 hyperfine state was modified. Figure 3.8 shows in red the frequency detunings for adressing the hyperfine state $^3P_1, F = 11/2$ used on the red MOT stages and the optical pumping. The beam for these two is tuned via a first double pass AOM called "Inj slave 1" at center frequency $f_{InjSlave1} = \pm 300$MHz and a second one at $f_{11/2} = \pm 200$: "AOM PompOpt" in a case, "AOM MOT 87" in the other case, both in double pass configuration.

3.4.1 Technical issues of inhabilitation of photoassociation

3.4.1.1 Laser width

3.5 87Sr molecules

Lopt questions sur nb quantique / choix de pompage optique

3.5.1 Physical sources of inhabilitation of photoassociation

3.5.1.1 On F = 9/2 : predissociation

3.5.1.2 Coupling to more energetic state from the IR

3.5.1.3 Node of wavefunction for some vibrational states

3.5.2 Energy landscape of 87Sr-87Sr molecules

Conclusion

Bibliography

[1] L. Pezze' and A. Smerzi, "Quantum theory of phase estimation," Nov. 2014. arXiv:1411.5164 [quant-ph].

[2] J. Dalibard, "Les interactions entre atomes dans les gaz quantiques,"

[3] J. H. Han, J. H. Kang, M. Lee, and Y.-i. Shin, "Photoassociation spectroscopy of ultracold ^{173}Yb atoms near the intercombination line," *Physical Review A*, vol. 97, p. 013401, Jan. 2018. arXiv:1710.03072 [physics].

[4] T. Zelevinsky, M. M. Boyd, A. D. Ludlow, T. Ido, J. Ye, R. Ciurylo, P. Naidon, and P. S. Julienne, "Narrow Line Photoassociation in an Optical Lattice," *Physical Review Letters*, vol. 96, p. 203201, May 2006. Number: 20 arXiv:physics/0602135.

[5] B. J. Reschovsky, B. P. Ruzic, H. Miyake, N. C. Pisenti, P. S. Julienne, and G. K. Campbell, "Narrow-line photoassociation spectroscopy and mass-scaling of bosonic strontium," Oct. 2018. arXiv:1808.06507 [physics].

[6] M. Borkowski, P. Morzyński, R. Ciuryło, P. S. Julienne, M. Yan, B. J. DeSalvo, and T. C. Killian, "Mass scaling and nonadiabatic effects in photoassociation spectroscopy of ultracold strontium atoms," *Physical Review A*, vol. 90, p. 032713, Sept. 2014.

Appendix A

Algorithms